

Microeconomics

Lecture 5: Elasticity of Demand

Marco Rosso

University of Bologna

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What we know so far

- Derivation of the demand function: $x^D(p_x, p_y, M)$
- Substitution effect and income effect
- Normal goods vs inferior goods
- Giffen goods (very rare cases)

Next step

How to **measure** the responsiveness of demand to changes in price, income and the prices of other goods.

Learning Objectives

After this lecture you will be able to:

1. **Compute** the income effect and the substitution effect
2. **Compute** the price elasticity of demand
3. **Interpret** its values (elastic/inelastic/unit elastic)
4. **Apply** the relationship between elasticity and total revenue
5. **Distinguish** income and cross-price elasticity
6. **Use** elasticity for business decisions

Focus of the Lecture

Understand how to use the elasticity of demand

Slutsky Equation

$$\frac{\partial x^D}{\partial p_x} = \underbrace{\frac{\partial x^c}{\partial p_x}}_{\text{Substitution effect}} - \underbrace{x^D \frac{\partial x^D}{\partial M}}_{\text{Income effect}}$$

Interpretation:

- $\frac{\partial x^c}{\partial p_x}$: substitution effect (always ≤ 0)
- $\frac{\partial x^D}{\partial M}$:
 - > 0 normal good
 - < 0 inferior good
- x^D : the income effect is proportional to the quantity consumed

Exercise 1: Slutsky Decomposition

Marshallian demand:

$$x^D(p, M) = 20 - 2p + 0.1M$$

Initial data:

$$p = 5, \quad M = 100$$

1. Compute x^D
2. Compute $\frac{\partial x^D}{\partial p}$
3. Compute $\frac{\partial x^D}{\partial M}$
4. Decompose the price effect using Slutsky

Solution

$$x^D = 20 - 2(5) + 0.1(100) = 20 - 10 + 10 = 20$$

$$\frac{\partial x^D}{\partial p} = -2 \quad \frac{\partial x^D}{\partial M} = 0.1$$

Slutsky:

$$\frac{\partial x^D}{\partial p} = \frac{\partial x^c}{\partial p} - x^D \frac{\partial x^D}{\partial M}$$

$$-2 = \frac{\partial x^c}{\partial p} - 20(0.1)$$

$$-2 = \frac{\partial x^c}{\partial p} - 2$$

$$\Rightarrow \frac{\partial x^c}{\partial p} = 0$$

Interpretation: The price effect is entirely due to the income effect.

Exercise 2: Inferior Good

$$x^D(p, M) = 30 - p - 0.05M$$

Data:

$$p = 10, \quad M = 200$$

1. Compute x^D
2. Compute the total price effect
3. Decompose it into substitution and income effects

Solution

$$x^D = 30 - 10 - 0.05(200) = 30 - 10 - 10 = 10$$

$$\frac{\partial x^D}{\partial p} = -1 \quad \frac{\partial x^D}{\partial M} = -0.05$$

Slutsky:

$$-1 = \frac{\partial x^c}{\partial p} - 10(-0.05)$$

$$-1 = \frac{\partial x^c}{\partial p} + 0.5$$

$$\frac{\partial x^c}{\partial p} = -1.5$$

Interpretation:

- Substitution effect: -1.5
- Income effect: +0.5
- Total: -1

Exercise 3: From a Derivative to a Percentage Change

Let us go back to Exercise 1:

$$x^D = 20 \quad \frac{\partial x^D}{\partial p} = -2 \quad p = 5$$

Question: If p increases by 1%, by how much (in %) does x change?

Solution

If p increases by 1%, then $\Delta p \approx 0.01 p$.

$$\Delta x \approx \frac{\partial x^D}{\partial p} \Delta p = (-2) (0.01 \cdot 5) = -0.1$$

In percentage terms:

$$\frac{\Delta x}{x} \approx \frac{-0.1}{20} = -0.005 = -0.5\%$$

Message

A 1% increase in price reduces the quantity by about **0.5%**. (This number is exactly the elasticity.)

Why do we need elasticity?

With Slutsky we computed a **sensitivity**:

$$\frac{\partial x^D}{\partial p_x}$$

which tells us how much x changes if p_x increases by **1 unit**.

Problem: this measure depends on the **units of measurement**

- liters vs bottles $\Rightarrow$ the derivative changes
- it is not comparable across different goods/markets

Key question

“If the price increases by 1%, by how much (in %) does the quantity change?”

Definition

Elasticity measures by how much (in %) the quantity changes when the price changes by 1%.

Price Elasticity: Intuition

Formula:

$$\varepsilon_P = \frac{\partial x^D}{\partial p_x} \cdot \frac{p_x}{x^D}$$

Interpretation:

- $\varepsilon_P = -0.5 \Rightarrow$ if p increases by 1%, x decreases by 0.5%
- for downward-sloping demand: $\varepsilon_P < 0$ (we often use $|\varepsilon_P|$)
- In practice we often look at $|\varepsilon_P|$ to classify (elastic/inelastic)

Rule of thumb

$$\Delta x\% \approx \varepsilon_P \cdot \Delta p\%$$

Elasticity and Slutsky: what is inside ε_P ?

From Slutsky:

$$\frac{\partial x^D}{\partial p_x} = \underbrace{\frac{\partial x^c}{\partial p_x}}_{\text{substitution}} - \underbrace{x^D \frac{\partial x^D}{\partial M}}_{\text{income}}$$

Multiplying by $\frac{p_x}{x^D}$ we obtain:

$$\varepsilon_P = \left(\frac{\partial x^c}{\partial p_x} \right) \frac{p_x}{x^D} - \left(\frac{\partial x^D}{\partial M} \right) p_x$$

Message

Price elasticity combines:

- the **substitution effect** (always ≤ 0)
- the **income effect** (it depends: normal vs inferior good)

Classification: Elastic vs Inelastic

Case	Value	Interpretation
Perfectly inelastic	$ \varepsilon = 0$	Q does not change as P varies (vertical)
Inelastic	$ \varepsilon < 1$	Q changes less than proportionally
Unit elastic	$ \varepsilon = 1$	Q changes exactly in proportion to P
Elastic	$ \varepsilon > 1$	Q changes more than proportionally
Perfectly elastic	$ \varepsilon = \infty$	Q drops to zero if P increases (horizontal)

Example: Linear Demand

Demand function: $Q = 100 - 2P$

Step 1: Compute the derivative

$$\frac{dQ}{dP} = -2$$

Step 2: Point elasticity

$$\varepsilon = -2 \cdot \frac{P}{Q}$$

Step 3: Evaluate at a specific point $P = 10$

$$Q = 100 - 2(10) = 80$$

$$\varepsilon = -2 \cdot \frac{10}{80} = -0.25 \quad \Rightarrow \quad |\varepsilon| = 0.25$$

Interpretation: At the price $P=10$, demand is **inelastic** ($|\varepsilon| < 1$):

If P increases by 1%, Q decreases by only 0.25%.

Example: Computation at a Different Point

Same function: $Q = 100 - 2P$, $\varepsilon = -2 \cdot \frac{P}{Q}$

Point A: $P = 40$

$$Q = 100 - 2(40) = 20$$

$$\varepsilon = -2 \cdot \frac{40}{20} = -4 \quad \Rightarrow \quad |\varepsilon| = 4$$

Elastic demand ($|\varepsilon| > 1$)

Point B: $P = 25$ (midpoint)

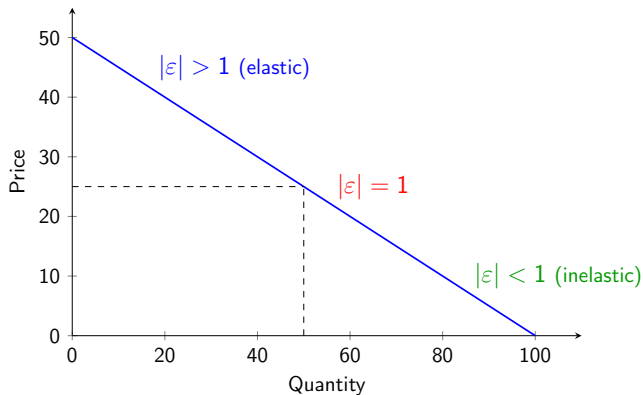
$$Q = 100 - 2(25) = 50$$

$$\varepsilon = -2 \cdot \frac{25}{50} = -1 \quad \Rightarrow \quad |\varepsilon| = 1$$

Unit elasticity

Important remark: Elasticity **varies** along the curve!

Graph: Elasticity Along a Linear Demand



Rule for linear demand:

- High prices (low Q): **elastic** demand
- Midpoint: **unit** elasticity
- Low prices (high Q): **inelastic** demand

Determinants of Elasticity

1. Availability of substitutes

- More substitutes $\Rightarrow$ more elastic (easy to switch)
- E.g.: Coca-Cola (many substitutes) vs insulin (no substitutes)

2. Necessity vs luxury

- Necessity $\Rightarrow$ inelastic (I have to buy it anyway)
- Luxury $\Rightarrow$ elastic (I can postpone)

3. Share of the budget

- Large % of income $\Rightarrow$ more elastic
- E.g.: car (elastic) vs salt (inelastic)

4. Time horizon

- Short run $\Rightarrow$ inelastic (habits, contracts)
- Long run $\Rightarrow$ elastic (time to adjust)

Real-World Examples of Elasticity

Good	$ \varepsilon $	Type
Insulin	0.1	Very inelastic
Salt	0.1	Very inelastic
Gasoline (short run)	0.2	Inelastic
Coffee	0.3	Inelastic
Clothing	0.9	Nearly unit elastic
Cars	1.2	Elastic
Restaurants	2.3	Very elastic
Air travel	2.4	Very elastic

Remark: Essential goods tend to be inelastic, luxury goods tend to be elastic.

Exercise 1

Demand function: $Q = 50 - 0.5P$

Questions:

1. Compute the elasticity at the point $P = 20$
2. Classify demand at that point
3. If P increases by 5%, by how much does Q change?

Solution:

1. $\frac{dQ}{dP} = -0.5$; $Q(20) = 50 - 0.5(20) = 40$
 $\varepsilon = -0.5 \cdot \frac{20}{40} = -0.25 \Rightarrow |\varepsilon| = 0.25$
2. **Inelastic** demand ($|\varepsilon| < 1$)
3. $\Delta Q\% = \varepsilon \cdot \Delta P\% = -0.25 \times 5\% = -1.25\% \rightarrow Q$ decreases by 1.25%

Exercise 2

Empirical data:

- Initial price: $P_1 = 10\text{€}$, Quantity $Q_1 = 200$ units
- Final price: $P_2 = 12\text{€}$, Quantity $Q_2 = 160$ units

Compute the elasticity:

$$\varepsilon = \frac{\Delta Q / Q_{\text{avg}}}{\Delta P / P_{\text{avg}}}$$

Solution:

$$Q_{\text{avg}} = \frac{200 + 160}{2} = 180, \quad P_{\text{avg}} = \frac{10 + 12}{2} = 11$$

$$\Delta Q = 160 - 200 = -40, \quad \Delta P = 12 - 10 = 2$$

$$\varepsilon = \frac{-40/180}{2/11} = \frac{-0.222}{0.182} \approx -1.22$$

$|\varepsilon| = 1.22 > 1 \Rightarrow$ **Elastic** demand

Summary of Part 1

Key concepts:

- Elasticity = percentage measure of responsiveness
- $\varepsilon = \frac{dQ}{dP} \cdot \frac{P}{Q}$
- $|\varepsilon| > 1$: elastic, $|\varepsilon| < 1$: inelastic
- Elasticity varies along the curve (linear demand)
- Determinants: substitutes, necessity, budget share, time

Elasticity and Total Revenue

Definition:

$$TR = P \times Q$$

Crucial question for firms:

"If I raise the price, does total revenue increase or decrease?"

Two opposing effects:

- **Price effect:** $P \uparrow \Rightarrow$ I earn more per unit
- **Quantity effect:** $Q \downarrow \Rightarrow$ I sell fewer units

Which effect dominates? It depends on the **elasticity!**

Fundamental Rule: Elasticity and Revenue

Elasticity–Revenue Relationship

- If demand is **elastic** ($|\varepsilon| > 1$): $P \uparrow \Rightarrow TR \downarrow$ (quantity effect dominates)
- If demand is **inelastic** ($|\varepsilon| < 1$): $P \uparrow \Rightarrow TR \uparrow$ (price effect dominates)
- If elasticity is **unitary** ($|\varepsilon| = 1$): $P \uparrow \Rightarrow TR$ constant (effects offset each other)

Intuition:

- Elastic: very price-sensitive customers $\Rightarrow$ I lose too many customers
- Inelastic: price-insensitive customers $\Rightarrow$ I lose few customers

Mathematical Proof

Differentiate TR with respect to P :

$$\frac{dTR}{dP} = \frac{d(P \cdot Q)}{dP} = Q + P \cdot \frac{dQ}{dP}$$

Divide by Q :

$$\frac{1}{Q} \cdot \frac{dTR}{dP} = 1 + \frac{P}{Q} \cdot \frac{dQ}{dP} = 1 + \varepsilon$$

Therefore:

$$\frac{dTR}{dP} = Q(1 + \varepsilon)$$

Conclusions:

- If $|\varepsilon| > 1 \Rightarrow \varepsilon < -1 \Rightarrow 1 + \varepsilon < 0 \Rightarrow \frac{dTR}{dP} < 0$
- If $|\varepsilon| < 1 \Rightarrow -1 < \varepsilon < 0 \Rightarrow 1 + \varepsilon > 0 \Rightarrow \frac{dTR}{dP} > 0$

Example: Gasoline

Context: Gasoline demand is inelastic in the short run ($|\varepsilon| \approx 0.3$)

Scenario: An oil company raises its price by 10%

Effect on quantity:

$$\Delta Q\% = \varepsilon \times \Delta P\% = -0.3 \times 10\% = -3\%$$

Effect on revenue:

$$TR_{\text{new}} \approx TR_{\text{old}} \times (1 + 0.10) \times (1 - 0.03) \approx TR_{\text{old}} \times 1.067$$

Result: Revenue increases by **6.7%**!

Managerial Implication

With inelastic demand, the firm has **pricing power**: it can raise prices without losing much volume.

Example: Luxury Restaurants

Context: Restaurant demand is elastic ($|\varepsilon| \approx 2.3$)

Scenario: A restaurant raises its prices by 10%

Effect on quantity:

$$\Delta Q\% = -2.3 \times 10\% = -23\%$$

Effect on revenue:

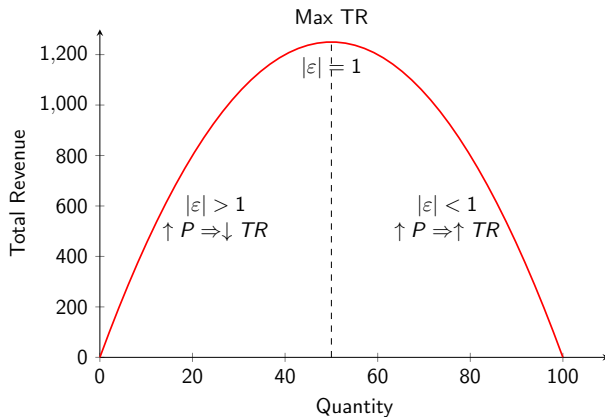
$$TR_{\text{new}} \approx TR_{\text{old}} \times 1.10 \times 0.77 \approx TR_{\text{old}} \times 0.847$$

Result: Revenue **decreases** by 15.3%!

Managerial Implication

With elastic demand, raising prices is **counterproductive**. Better strategy: cut prices to increase volume and revenue.

Total Revenue Curve: Linear Demand



Remarks:

- TR is maximized when $|\epsilon| = 1$
- Before the max: elastic demand (better to lower P)
- After the max: inelastic demand (better to raise P)

Optimal Pricing Strategy

Business decision:

If demand is...	To $\uparrow$ TR	Action
Elastic ($ \varepsilon > 1$)	You need more customers	Cut P
Unit elastic ($ \varepsilon = 1$)	You are already at the optimum	Do not change
Inelastic ($ \varepsilon < 1$)	Squeeze loyal customers	Raise P

Important caveat:

- This is a strategy to **maximize revenue**, not profit!
- Profit also depends on **costs** (we will see producer theory)
- But it is a very good approximation for short-run decisions

Application: Price Discrimination

Idea: Market segments with different elasticities $\Rightarrow$ different prices

Example: Airline Tickets

- **Business travelers:** inelastic demand ($|\varepsilon| \approx 0.5$) $\Rightarrow$ High price (€500+)
- **Tourists:** elastic demand ($|\varepsilon| \approx 3$) $\Rightarrow$ Low price (€50–100, booked in advance)

Example: Cinema

- **Adults:** $|\varepsilon| \approx 1 \Rightarrow$ Standard price (€10)
- **Students/seniors:** $|\varepsilon| > 1 \Rightarrow$ Discount (€7)

Logic: Extract **consumer surplus** by differentiating prices according to elasticity!

Exercise: Maximizing Revenue

Demand function: $Q = 120 - 3P$

Question: At what price is revenue maximized?

Solution, Method 1 (elasticity):

Revenue is maximized when $|\varepsilon| = 1$:

$$\left| -3 \cdot \frac{P}{Q} \right| = 1 \quad \Rightarrow \quad 3P = Q$$

Substitute $Q = 120 - 3P$:

$$3P = 120 - 3P \quad \Rightarrow \quad 6P = 120 \quad \Rightarrow \quad P^* = 20$$

$$Q^* = 120 - 3(20) = 60, \quad TR^* = 20 \times 60 = 1,200$$

Solution, Method 2 (derivative):

$$TR = P \cdot Q = P(120 - 3P) = 120P - 3P^2$$

$$\frac{dTR}{dP} = 120 - 6P = 0 \quad \Rightarrow \quad P^* = 20 \quad \checkmark$$

Business Case: SuperMercato SpA

The Situation

SuperMercato SpA has just completed a market study on its private-label pasta. The results show a price elasticity of demand equal to **-0.4**.

The current price is **€1.20 per pack** and they sell **50,000 packs per month**.

The Question

What should the company do to increase revenue?

Group Activity

Form groups of 3–4 people and discuss the following questions:

1. Is demand elastic or inelastic?
2. If they raise the price by 10%, what happens to revenue?
3. Which pricing strategy would you recommend?
4. What other factors should the company consider?

Hint: Compute the quantitative impact using the elasticity formula

Solution: Elasticity Analysis

1. Classification: $\varepsilon = -0.4 \Rightarrow |\varepsilon| = 0.4 < 1$

$\Rightarrow$ INELASTIC demand

2. Interpretation:

- If P increases by 1%, Q decreases by only 0.4%
- Consumers are not very price-sensitive
- Pasta is a staple good

Solution: Computing the Price Impact

Scenario: 10% price increase (from €1.20 to €1.32)

Change in quantity:

$$\Delta Q\% = \varepsilon \times \Delta P\% = -0.4 \times 10\% = -4\%$$

New quantity:

$$Q_{\text{new}} = 50,000 \times (1 - 0.04) = 48,000 \text{ packs}$$

Solution: Impact on Revenue

	Initial	New
Price	€1.20	€1.32
Quantity	50,000	48,000
Total Revenue	€60,000	€63,360

Result $\Rightarrow$ Revenue increase: +€3,360 (+5.6%)

Strategic Recommendation

Raise the Price

With inelastic demand ($|\varepsilon| = 0.4$), a price increase raises total revenue.

Advantages:

- +5.6% revenue with a limited loss of volume (-4%)
- Higher margin per unit sold
- Possible improvement in profit (if fixed/variable costs are favorable)

Watch out for:

- Competitors' reaction (price war?)
- Elasticity might change in the long run
- Perceived quality vs value for money

Other Factors to Consider

Competition

- Cross-price elasticity with rival brands?
- Risk of customers switching?

Brand Image

- Do high prices signal quality?
- Or do they drive away price-sensitive segments?

Costs

- Does selling less reduce costs?
- Impact on net profit?

Long Run

- Elasticity tends to increase
- Monitor it constantly!

Key Takeaways from the Case

1. Inelastic demand $\Rightarrow$ raising the price increases revenue
2. **Always** compute the quantitative impact before deciding
3. Consider cross-price elasticity (competition)
4. Monitor elasticity over time (it can change)
5. Revenue $\neq$ Profit (costs are needed too!)

Summary of Part 2

Key concepts:

- $TR = P \times Q$: price effect vs quantity effect
- Elastic: $P \uparrow \Rightarrow TR \downarrow$ (cut price to $\uparrow$ revenue)
- Inelastic: $P \uparrow \Rightarrow TR \uparrow$ (raise price to $\uparrow$ revenue)
- TR is maximized when $|\varepsilon| = 1$
- Price discrimination exploits differences in elasticity

Income Elasticity

Definition

The **income elasticity of demand** measures the percentage change in quantity demanded in response to a percentage change in income.

Formula:

$$\varepsilon_M = \frac{\Delta Q/Q}{\Delta M/M} = \frac{\partial Q}{\partial M} \cdot \frac{M}{Q}$$

Interpretation:

- $\varepsilon_M > 0$: normal good (demand rises with income)
- $\varepsilon_M < 0$: inferior good (demand falls with income)

Classification by Income Elasticity

Type	ε_M	Examples
Inferior good	< 0	Public transport, cheap pasta, second-hand clothing
Necessity (normal)	$0 < \varepsilon_M < 1$	Bread, electricity, basic health services
Unit elastic (normal)	$\varepsilon_M \approx 1$	Clothing, furniture, household appliances
Luxury good (normal)	$\varepsilon_M > 1$	Travel, restaurants, luxury cars, jewelry

Implications:

- During recessions: demand for luxury goods collapses
- During economic growth: demand for inferior goods falls

Example: Income Elasticity

Demand function: $Q = 0.02M - 5P$ (M = income)

Data: $M = 5,000$, $P = 10$

Step 1: Compute the quantity

$$Q = 0.02(5,000) - 5(10) = 100 - 50 = 50$$

Step 2: Compute the income elasticity

$$\begin{aligned}\frac{\partial Q}{\partial M} &= 0.02 \\ \varepsilon_M &= 0.02 \cdot \frac{5000}{50} = 0.02 \times 100 = 2\end{aligned}$$

Interpretation: $\varepsilon_M = 2 > 1 \Rightarrow$ **Luxury good**

If income increases by 10%, demand increases by 20%.

Cross-Price Elasticity

Definition

The **cross-price elasticity** between goods x and y measures the percentage change in the demand for x in response to a percentage change in the price of y .

Formula:

$$\varepsilon_{xy} = \frac{\Delta Q_x / Q_x}{\Delta P_y / P_y} = \frac{\partial Q_x}{\partial P_y} \cdot \frac{P_y}{Q_x}$$

Classification:

- $\varepsilon_{xy} > 0$: **substitute** goods (Coke vs Pepsi)
- $\varepsilon_{xy} < 0$: **complementary** goods (cars and gasoline)
- $\varepsilon_{xy} = 0$: **independent** goods

Interpreting Cross-Price Elasticity

Substitute Goods ($\varepsilon_{xy} > 0$):

- $P_y \uparrow \Rightarrow$ consumers switch to $x \Rightarrow Q_x \uparrow$
- Example: the price of butter rises $\Rightarrow$ demand for margarine rises

Complementary Goods ($\varepsilon_{xy} < 0$):

- $P_y \uparrow \Rightarrow$ they consume less $y \Rightarrow$ less x is needed $\Rightarrow Q_x \downarrow$
- Example: the price of cars rises $\Rightarrow$ demand for gasoline falls

Strength of the relationship:

- Large $|\varepsilon_{xy}|$: **close** substitutes/complements
- Small $|\varepsilon_{xy}|$: **weak** relationship

Examples of Cross-Price Elasticity

Pair of goods	ε_{xy}	Type
Butter - Margarine	+0.81	Close substitutes
Coca-Cola - Pepsi	+0.63	Substitutes
Gasoline - Cars	-0.28	Complements
Coffee - Sugar	-0.15	Weak complements
Bread - Meat	+0.05	Nearly independent

Applications:

- Forecasting the impact of competitors' prices on own sales
- Decisions on bundles of complementary products
- Antitrust analysis (relevant market definition)

Exercise: Three Elasticities

Demand for cars:

$$Q = 100 + 0.01M - 2P_{\text{car}} - 0.5P_{\text{gas}}$$

Note: P_{car} is measured in thousands of euros.

Current values:

$$M = 50,000 \quad P_{\text{car}} = 20 \quad P_{\text{gas}} = 2$$

Compute:

1. Quantity demanded Q
2. Price elasticity ε_P
3. Income elasticity ε_M
4. Cross-price elasticity with respect to gasoline ε_{gas}
5. Interpret the results

Solution (1)

1. Quantity:

$$Q = 100 + 0.01(50,000) - 2(20) - 0.5(2)$$
$$Q = 100 + 500 - 40 - 1 = 559$$

2. Price elasticity:

$$\frac{\partial Q}{\partial P_{\text{car}}} = -2$$
$$\varepsilon_P = -2 \cdot \frac{20}{559} = -0.072$$
$$|\varepsilon_P| < 1 \Rightarrow \textbf{Inelastic demand}$$

3. Income elasticity:

$$\frac{\partial Q}{\partial M} = 0.01$$
$$\varepsilon_M = 0.01 \cdot \frac{50,000}{559} = 0.89$$
$$0 < \varepsilon_M < 1 \Rightarrow \textbf{Normal good (necessity)}$$

Solution (2)

4. Cross-price elasticity (gasoline):

$$\frac{\partial Q}{\partial P_{\text{gas}}} = -0.5$$

$$\varepsilon_{\text{gas}} = -0.5 \cdot \frac{2}{559} = -0.0018$$

$\varepsilon < 0 \Rightarrow$ **Complementary goods (weak)**

5. Interpretation of the results

- **Price elasticity** $\varepsilon_P = -0.072 \Rightarrow$ **highly inelastic** demand: if P_{car} increases by 1%, Q decreases by about **0.072%** (weak price effect).
 - **Income elasticity** $\varepsilon_M = 0.89 \Rightarrow$ **normal good (necessity)**: if M increases by 1%, Q increases by about **0.89%** (almost proportional).
 - **Cross-price elasticity** $\varepsilon_{\text{gas}} = -0.0018 \Rightarrow$ **complementarity**: if P_{gas} increases by 1%, Q decreases by about **0.0018%**. **Very weak** relationship (almost zero impact).
- Demand responds very little to prices, but fairly strongly to income.

Overview: All Elasticities

Type	Formula	Measures
Price elasticity	$\varepsilon_P = \frac{\partial Q}{\partial P} \cdot \frac{P}{Q}$	Responsiveness to own price
Income elasticity	$\varepsilon_M = \frac{\partial Q}{\partial M} \cdot \frac{M}{Q}$	Responsiveness to income
Cross-price elasticity	$\varepsilon_{xy} = \frac{\partial Q_x}{\partial P_y} \cdot \frac{P_y}{Q_x}$	Relationship between goods

They all share:

- **Percentage** measures (unit-free)
- General formula: (derivative) $\times$ (ratio of variables)
- They allow comparisons across different markets

Managerial and Policy Applications

1. Pricing Strategy

- Elastic demand? Cut prices to gain market share
- Inelastic demand? Raise prices to extract surplus

2. Sales Forecasting

- $$\Delta Q = \varepsilon_P \cdot \frac{\Delta P}{P} \cdot Q + \varepsilon_M \cdot \frac{\Delta M}{M} \cdot Q$$

3. Fiscal Policy

- Taxing inelastic goods (cigarettes, gasoline) raises more revenue
- Subsidizing goods with high income elasticity supports growth

4. Competitive Analysis

- High cross-price elasticity? Very competitive market
- Low? You have market power (product differentiation)

Overall Summary

Part 1: Price Elasticity

- Formula: $\varepsilon = \frac{dQ}{dP} \cdot \frac{P}{Q}$
- Classification: elastic (> 1), inelastic (< 1), unit elastic ($= 1$)
- Determinants: substitutes, necessity, budget share, time

Part 2: Elasticity and Revenue

- Elastic: $P \uparrow \Rightarrow TR \downarrow$
- Inelastic: $P \uparrow \Rightarrow TR \uparrow$
- TR is maximized when $|\varepsilon| = 1$

Part 3: Other Elasticities

- Income elasticity: ε_M (normal vs inferior, luxury)
- Cross-price elasticity: ε_{xy} (substitutes vs complements)

Skills Acquired

- ✓ **Compute** elasticity from a demand function or empirical data
- ✓ **Classify** goods by elasticity (elastic/inelastic)
- ✓ **Predict** the impact of price changes on total revenue
- ✓ **Apply** income elasticity to business-cycle forecasts
- ✓ **Identify** relationships between goods using cross-price elasticity
- ✓ **Recommend** pricing strategies based on elasticity

Optional Homework Exercises

1. Demand: $Q = 200 - 5P$. Compute $|\varepsilon|$ at $P = 10$ and $P = 30$.
2. If $|\varepsilon| = 1.5$ and P increases by 4%, by how much does Q change?
3. $TR = 100P - 2P^2$. At what price is TR maximized?
4. Function: $Q = 10 + 0.02M - 3P$. Compute ε_P and ε_M at $M = 5,000$, $P = 20$.
5. Two goods with $\varepsilon_{xy} = 0.8$. If P_y increases by 5%, by how much does Q_x change? Are they substitutes or complements?